

Evidence-based Decision-making

Evidence-based practice (EBP) is a method used to apply critical thinking to the best available evidence when making decisions.¹ EBP developed in the field of medicine towards the end of the 20th century² as a way to help improve treatment and patient outcomes. It addressed the problem that medical practitioners didn't always have access to the latest research, which sometimes led them to rely too heavily on their own judgement when making decisions.

Evidence-based practice has since been implemented in many other disciplines – including management.³ As the name implies, the core element of the practice is evidence. But what qualifies as evidence? Put simply, evidence is anything that can be presented as a reliable basis upon which to reach a decision. It's made up of various types of data and information – including scientific research, facts and figures, as well as expertise gained through experience.

For example, let's say that you're aiming to improve operational efficiency within your department. To gather evidence for this task, you might start by analysing workflow data to identify patterns in process bottlenecks. You could conduct one-to-one meetings with team members to understand their challenges and gather insights about the tools and processes they find most effective. You might also review academic papers or research good practice in operational planning from similar organisations in your industry. By combining these insights, you'll have a solid 'evidence base' from which to guide strategies to enhance your department's operational efficiency.

Everyone uses evidence to some extent in their decision-making. However, they might not use enough evidence, or consider whether it is strong enough. For instance, people sometimes apply their personal experience without taking into account that it is unlikely to be universally applicable.

As an evidence-based practitioner, you can avoid this by being **conscientious**, **explicit**, and **judicious** when handling data and information.⁴

- **Conscientious** – put in the time and effort to find the most complete data and information to understand a situation
- **Explicit** – be clear about how you source and describe the evidence you collect, so that others can evaluate it
- **Judicious** – appraise the quality of the evidence and only select the most appropriate data and information

With this approach, you can make well-supported decisions that are more likely to work.

¹ CIPD (2024). Evidence-based practice for effective decision-making. Factsheet. Available at: <https://www.cipd.co.uk/knowledge/strategy/analytics/evidence-based-practice-factsheet> [Accessed 28 October 2024]

² Thoma, A. and Eaves, F. F. (2015). A brief history of evidence-based medicine (EBM) and the contributions of Dr David Sackett. *Aesthetic Surgery Journal*, 35(8), pp. 261–263. Available at: <https://academic.oup.com/asi/article/35/8/NP261/251339> [Accessed 28 October 2024]

³ CEBMa (N.D.). What is evidence-based management? Available at: <https://cebma.org/resources/frequently-asked-questions/what-is-evidence-based-management> [Accessed 28 October 2024]

⁴ Briner, R. (2019). The basics of evidence-based practice. HR People + Strategy. SHRM Executive Network. Available at: <https://cebma.org/assets/Uploads/Briner-The-Basics-of-Evidence-Based-Practice.pdf> [Accessed 28 October 2024]

As a manager making important decisions, you also have a moral obligation to use the best available evidence. This will lead to better outcomes and demonstrate your effectiveness in your role. After all, you want to show that you are making decisions and giving guidance based on what the evidence shows will work.

The ultimate aim of evidence-based practice in management is to help managers and leaders make better decisions. These can, in turn, lead to more effective strategies, improved team performance and stronger business outcomes. By relying on a solid foundation of evidence – rather than intuition or assumptions – managers can address challenges more accurately, reduce risks, and make choices that are more likely to achieve team and organisational goals.

Types of evidence

The evidence that managers use to support their decisions is made up of **data** and **information**. While these terms are often used interchangeably, they represent two distinct concepts.

Data is made up of basic values – such as facts, figures, or opinions – which have been collected but not manipulated in any way. This is why data is often described as ‘raw’. In an operational planning context, data might include metrics like production times, inventory levels or customer order volumes.

Information, on the other hand, is data that has been organised and analysed. Information may include statistics, which can show patterns in data, or reports that put them into a wider context.

While raw data is valuable, transforming it into information can give it more meaning, relevance, and purpose.⁵ For example, a manager might analyse production efficiency ratings by department, team or process. This could reveal trends, such as differences in efficiency across teams, which potentially signal operational weaknesses or areas needing more support.

But where can managers find data and information on which to base their decisions? Rob Briner of the Center for Evidence-Based Management (CEBMA), argues that we should rely on multiple sources, recommending the following four types of evidence:⁶

- **Scientific evidence:** Research findings that relate to the decision in question in some way. There is research available on a variety of work and management-related topics and most journals and studies can now be found online. Scientific databases such as ScienceDirect⁷ and Google Scholar⁸ provide either full texts or links to help you locate them. Not that full

⁵ Dykes, B. (2016). Actionable insights: The missing link between data and business value. Forbes. Available at: <https://www.forbes.com/sites/brentdykes/2016/04/26/actionable-insights-the-missing-link-between-data-and-business-value/?sh=a269c7151e57> [Accessed October 31 2024]

⁶ Briner, R. (2019). The basics of evidence-based practice. HR People + Strategy. SHRM Executive Network. Available at: <https://cebma.org/assets/Uploads/Briner-The-Basics-of-Evidence-Based-Practice.pdf> [Accessed 28 October 2024]

⁷ ScienceDirect (N.D.). Explore scientific, technical, and medical research on ScienceDirect. Available at: <https://www.sciencedirect.com/> [Accessed 31 October 2024]

⁸ Google (N.D.). Google Scholar. Available at: <https://scholar.google.com/> [Accessed 31 October 2024]

texts of academic publications are not always available free of charge.

- **Organisational data:** Data gathered within an organisation for its own purposes. For example, this could take the form of statistics on production rates, employee productivity or the results of operational audits.
- **Experiential data:** The professional input of managers, consultants, and other business experts – both within and outside a company. This is specialised knowledge acquired through experience. When this kind of data is gathered from people outside the organisation, there is often a charge to access it.
- **Stakeholder evidence:** The input of anyone who will be affected by a specific decision. This might be internal – for instance, the opinions of your team members and managers – or external, such as the views of customers and suppliers.

There are two important points to make about this list of evidence types. First, evidence is everywhere. Whatever the operational issue or problem you're trying to solve, there will be evidence available to help you reach a decision. The most relevant data is often found within your company but can still be supported by wider research. Second, it's unlikely that you will find good-quality evidence from all four sources every time. This is why it's important to be pragmatic and use the best available evidence, rather than waiting for the 'perfect' source to arrive.

Process for making evidence-based decisions

CEBMA suggests that managers should make evidence-based decisions according to the following process.⁹

1. **Asking:** Translating a practical issue or problem into an answerable question or hypothesis. For example, a production manager might ask, "What strategies can we implement to improve operational efficiency in our department?"
2. **Acquiring:** Searching for the evidence and retrieving it. In this case, the evidence might include operational data, process maps, employee feedback, market research and scientific research.
3. **Appraising:** Critically assessing the quality and relevance of the evidence – for example, the trustworthiness of methodologies used in research or how applicable evidence is to your specific organisational or team context.
4. **Aggregating:** Weighing and synthesising the data. Here, you'd ideally create a comprehensive overview of the most effective and relevant operational strategies based on the aggregated evidence.

⁹ Barends, E., Rousseau, D. M. and Briner, R. B. (2014). Evidence-based management: The basic principles. CEBMA. Available at: <https://cebma.org/assets/Uploads/Evidence-Based-Practice-The-Basic-Principles.pdf> [Accessed 31 October 2024]

5. **Applying:** Incorporating the evidence you've collated into your decision-making process. This involves choosing the operational strategies that have the strongest evidence base to support their effectiveness and tailoring them to fit the specific needs of your department.
6. **Assessing:** Evaluating the outcome of the decision you've made. For example, you might track changes in operational efficiency metrics over time and assess them to determine which strategies have had the greatest impact.